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Quantitative Research Z-score
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The following raw scores were obtained from the local administration of the Minnesota Management Skills Inventory. Possible scores on the test ranged from 0 to 100, with higher scores indicating outstanding decision-making skills. Find the z-score for each raw score. Use the method demonstrated in class or in the online study unit. Show all steps of your work.

Given:

Person	X
Bill	30
Bob	40
Anne	40
Amy	45
Joe	60

Set of scores = $X = \{30, 40, 40, 45, 60\}$

Number of scores = $N = 5$

1.1 Mean

Formula: $X' = \sum X / N$

Mean = $X' = \sum (30 + 40 + 40 + 45 + 60) / 5 = 215 / 5 = 43$

1.2 Median

This set of scores has an odd number of observations. The median is the middle score in the sorted list.

Median = 40

1.3 Mode

The mode is the most frequent score. For this set of scores, the mode is 40.

1.4 Variance

Formula: $V = \sum (X - X')^2 / N$

Person	X	(X-X')	(X-X')^2
Bill	30	-13	169
Bob	40	-3	9
Anne	40	-3	9
Amy	45	2	4
Joe	60	17	289

$$\text{Variance} = V = \sum (169 + 9 + 9 + 4 + 289) / 5 = 480 / 5 = 96$$

1.5 Standard Deviation

Formula : $SD = \sqrt{V}$

$$\text{Standard Deviation} = SD = \sqrt{96} = 9.80$$

2 Who got the highest raw score on the test?

From the set of scores, Joe's raw score is 60 and therefore he has the highest raw score.

3 In terms of interactive measurement, did that person do well on the test? (Very short answer.)

Joe's raw score is 60 out of a 100-point scale, or 60%. In term of performance, the letter grade for "60%" is "D". No, Joe did not do well.

4 What is that raw score's corresponding z score (standard score)?

Formula : $z \text{ score} = z = (X - X') / SD$

Person	X	(X-X')	(X-X')^2	(X-X') / SD
Bill	30	-13	169	-1.33
Bob	40	-3	9	-.31
Anne	40	-3	9	-.31
Amy	45	2	4	+.2
Joe	60	17	289	+1.73

Note: Individual z score for each value of X (or raw score).

5 What does that z score tell you about how that person did on the test?

Joe's z score is +1.73, and it indicates 1.73 deviations above the mean (average) compared to the other persons. Based on Joe's z score, he did well.

6 Find the z scores (standard scores) for each of the following raw scores obtained from a scale used to evaluate customer satisfaction in service at a national restaurant chain. Use the method demonstrated in class or in the online study unit.

Show all steps of your work.

Given:

Set of raw scores = $X = \{5, 6, 7, 8, 4\}$

Number of scores = $N = 5$

6.1 Without additional information, can you say which raw score and corresponding z score indicate the highest level of satisfaction?

Without additional information, we can't say that a low score of 4 or a high score of 8 indicates customer satisfaction. Furthermore, we don't know the scale of measurement.

Now, we need to gather more information about these raw scores to help us determine which raw score correlates with better customer satisfaction.

6.2 Starting with the mean

Formula: $X' = \sum X / N$

Mean = $X' = \sum (5 + 6 + 7 + 8 + 4) / 5 = 30 / 5 = 6$

6.3 Median

This set of scores has an odd number of observations; the median is the middle score in the sorted list.

X
4
5
6 ← median
7
8

Median = 40

6.4 Mode

There is no most frequent score, and therefore there is no mode.

6.5 Variance

Formula: $V = \sum (X - X')^2 / N$

X	(X-X')	(X-X') ²
4	-2	4
5	-1	1
6	0	0
7	1	1
8	2	4

$$\text{Variance} = V = \sum (4 + 1 + 0 + 1 + 4) / 5 = 10 / 5 = 2$$

6.6 Standard Deviation

Formula : $SD = \sqrt{V}$

$$\text{Standard Deviation} = SD = \sqrt{2} = 1.41$$

6.7 Z Score

Formula: $z \text{ score} = z = (X - X') / SD$

X	(X-X')	(X-X') ²	(X-X') / SD
4	-2	4	-1.42
5	-1	1	-.71
6	0	0	0
7	1	1	+.71
8	2	4	+1.42

6.8 Does the z score for the raw score of 6 have any meaning? If so, what is it?

Yes. The raw score of 6 is at the mean (average). Now, we can say that the raw score of 8 is the highest value of satisfaction because it is +1.42 above the mean. Based on the results, we can say that customers are neither satisfied nor dissatisfied.

- 7 Bob got a z score (standard score) of -2 on a completely different test. What can you say about the kind of test he was given, and how he performed on the test?

The assumption is that we compare Bob score from question 1 above.

From question:

Person	X	(X-X')	(X-X')^2	(X-X') / SD
Bill	30	-13	169	-1.33
Bob	40	-3	9	-.31
Anne	40	-3	9	-.31
Amy	45	2	4	+.2
Joe	60	17	289	+1.73

On a completely different kind of test and using the same raw scores for the other people

Bod	?	-18	?	-2
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$$\text{Z-score} = (X-X') / SD = -18/9 = -2$$

$$\text{Standard Deviation} = SD = \sqrt{81} = 9$$

$$\text{Variance} = SD^2 = 9 * 9 = 81$$

Hypothetically, Bob's new raw score is much less than his previous raw score of 30. A new z score of -2 is below the mean (or average). So, on a new, completely different test, Bob did much worse, or the new test is much harder than the previous one.

We don't know Bob's raw score or the other people, so we can't calculate the mean; therefore, we can't use interactive measurement either. All we know about Bob is his z-score, and based on it, he did not do well.

Here is the assumption:

Only one raw score = X = 32 or 32/100 or 30%

N = 1

Mean = 50 (the highest X score is 100)